# Observe, Imagine, Abstain: Detecting Hallucinated Geometry in Feed-Forward 3D Reconstruction

Anonymous CVPR submission

## Abstract

*Feed-forward 3D foundation models reconstruct a scene from unposed photographs in a single pass. Where no photograph shows a surface, they fill the gap from learned priors, and they often assign the invented geometry a confidence indistinguishable from that of observed geometry. We call this geometric hallucination and define it operationally: a predicted point is hallucinated when it lies on a surface that no input view observes. HalluBench-3D produces such labels for free by hiding a 3D region of the ground-truth surface in every input view, so that every pixel inside the region is unsupported and every pixel far from it is supported. We ask whether a model's native confidence separates the two (*AUROC*) and how much unsupported geometry survives a threshold that keeps 90% of supported geometry, the Confident-Unsupported Rate (*CUR*). Across five models (VGGT, Pi3, Pi3X, MapAnything, Depth Anything 3), 18 DTU scans and 18 CO3Dv2 scenes, native confidence lets between 24% and 81% of unsupported pixels through. A realistic inpainted fill makes leakage worse, not better; dropping views or adding a distractor image leaves it unchanged; and the cheapest alternative, variance over perturbed inputs, is worse than native confidence on both backbones we tried. The ranking of models also changes between the lab rig and phone captures: only Pi3 behaves the same on both. We then ask what a detector of unobserved geometry can be. A small* support head *on the frozen multi-view features of the backbone predicts, per pixel, whether the output has multi-view support, and abstains in between. It needs no ground truth: its labels are the support count of the backbone's own prediction, the quantity a reprojection test measures, and the head learns to read it from the features in one forward. On four backbones (VGGT, Pi3X, Pi3, Depth Anything 3) and four datasets its training never touched (DTU benchmark scans with 3 to 10 clean views, CO3Dv2 phone orbits, ETH3D building scans, Internet photographs of nine heritage landmarks) it separates single-view from supported geometry at* AUROC *0.93 to 0.99 against 0.55 to 0.89 for native confidence, matches a head trained on ground-truth visibility within 0.02, and beats a control that sees one image at a time, which falls below chance off the lab rig, as does the distance to the image border. It beats reprojection consistency on DTU, CO3Dv2 and the landmark photographs and ties it on ETH3D, at a fraction of the cost. Two cautions: a head trained on the benchmark's fills learns the fill, and no detector, this head included, fires on a filled patch that every view shows; five models of three families invent the same surface there, so disagreement between models does not find it either. Code, benchmark, heads and the full run log will be released.*

## 1. Introduction

Feed-forward 3D reconstruction has moved from pairwise pointmaps [14, 31] to models that take dozens of unposed images and return dense geometry in one forward pass [13, 17, 29, 32]. These models are trained to output geometry at every pixel, including pixels whose true surface is not visible in any input view. The training objective rewards a plausible guess, so the models learn to guess. The trouble is that the confidence map, the one signal a user has, does not say which parts were observed and which were guessed.

Suppose you are a conservator looking at a reconstruction of a dome, and you see a crack. Before you send someone up a scaffold, you want to know whether any photograph shows it or whether the model drew it from what domes usually look like. Today you cannot tell: the confidence map is bright in both cases.

We make three contributions.

1. **A definition and a benchmark.** We define hallucinated geometry as predicted points on surfaces that no input view shows. HalluBench-3D produces these labels by 3D-consistent masking of ground-truth surfaces, with controls on the same pixels under clean input. We report AUROC, CUR and CUR-ERR for five models on DTU and CO3Dv2, three fill types, an inpainted fill, and stress tests (view dropping, low parallax, a distractor image), each with cluster-bootstrap intervals and paired permutation tests.
2. **The backbone already knows, and a label-free way to read it.** Multi-view support is linearly decodable from the second half of the backbone's cross-view layers and

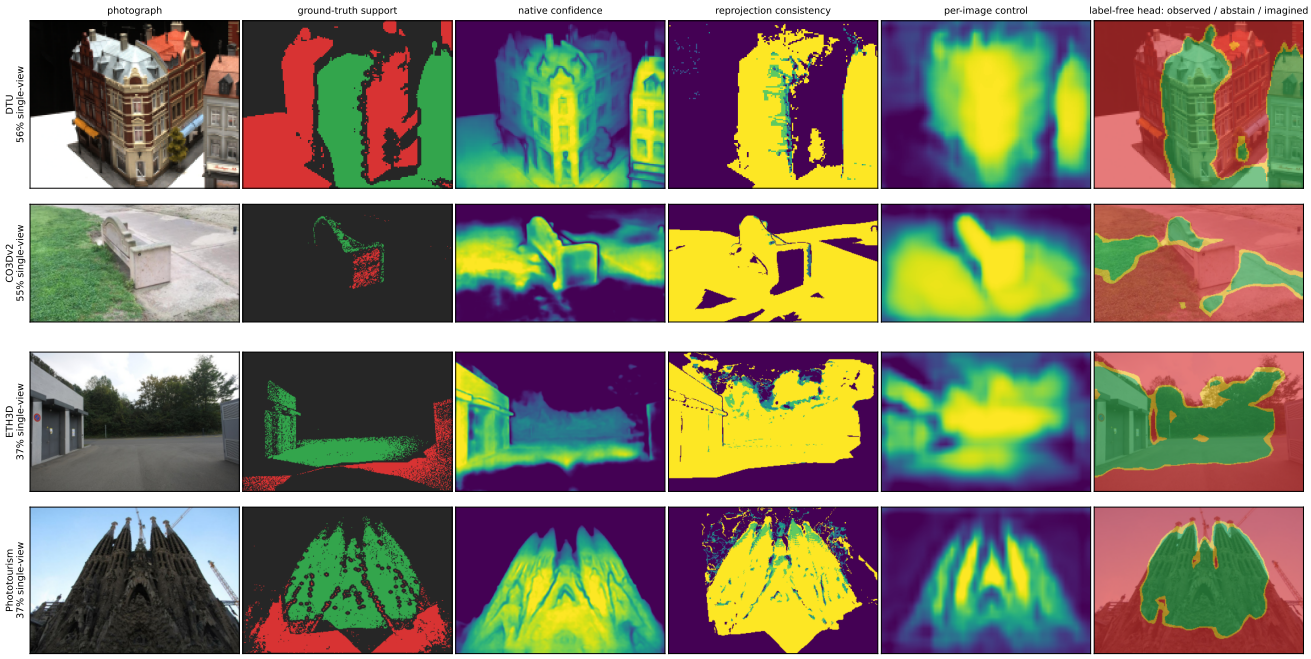



Figure 1. **What the label-free head says, on four captures it never saw.** One view of a clean multi-view input per dataset (DTU rig, CO3Dv2 phone orbit, ETH3D DSLR, a tourist photograph of the Sagrada Família); ground-truth support (red: no other input view sees the surface; green: two or more do; dark: no ground truth); VGGT's native confidence; reprojection consistency with its own cameras; the per-image control; and the head's bands at the fixed thresholds (green observed, yellow abstain, red imagined). The head's labels came from the backbone's own predictions on DTU training scans. Rendered from the released per-pixel scores (run P6-dump).

absent from its per-image tokens; a linear probe gets within 0.015 of any head we trained. We read that signal with a small head whose labels come from the backbone's own prediction, so it trains on any capture with no ground truth, and abstains in between. On four backbones and four datasets it never saw it passes two pre-registered tests against native confidence, a per-image control and a ground-truth-trained head (Section 7). Trained on the benchmark's synthetic fills instead, it learns the fill, which we report as a warning for masking-based supervision.

3. **Controls that a detector must pass, and one it fails.** Native confidence, perturbation variance, reprojection consistency, the distance to the image border, depth edges, a linear probe on the same features and a head that sees one image at a time are scored on identical rows with paired tests. Put back to the benchmark's own filled patches, the head does not fire: it certifies agreement between views, which a consistent invention has.

Every number comes from a logged run with a run id; the pass conditions of Section 7 were written down, with dates, before the heads they judge were trained (supplement).

The headline claim we test, and do not assume, is that native confidence fails to flag unsupported geometry across current models. It holds, and it holds more strongly on phone captures than on the lab rig.

## 2. Related work

**Feed-forward 3D reconstruction.** DUSt3R and MASt3R [14, 31] predict pointmaps for image pairs. VGGT [29], Pi3 and Pi3X [32], MapAnything [13], VGGT-Omega [30] and Depth Anything 3 [17] extend this to many unposed views in one pass. Each exposes a per-pixel confidence trained as a loss weight on observed data; none is trained to distinguish observed from unobserved surfaces. Pi3X replaces Pi3's near-binary confidence with continuous quality levels; MapAnything also offers a multi-view consistency confidence, which we include as a baseline.

**Hallucination in feed-forward 3D.** Bratulic et al. [3] probe DUSt3R, VGGT and DA3 and show that attention heads keep matching correspondences after small patches are occluded, concluding that these models hallucinate missing correspondences from learned priors; they measure attention and Sampson error on ShapeNet pairs, not confidence or 3D labels. Paul et al. [23] benchmark hallucination at the input level: unrelated scenes, repeated images and noise still produce dense geometry, scored by COLMAP-based consistency. 3D-Mirage [21] studies monocular depth under visual illusions. Li et al. [15] show that wide viewpoint spans produce unsupported geometry, and Dadajonov and Umarhojiyev [5] that leaving parts of a room unobserved

is not revealed by inference-time uncertainty; neither has per-point labels. FreeInpaint [22] fills masked regions on purpose, the mechanism these models apply unasked. Our definition is per point, with exact supported and unsupported labels, and it tests whether native confidence separates them.

**Uncertainty for feed-forward 3D.** ReVar3R [18] takes per-point variance over perturbed runs; it beats native confidence on AUSE in most conditions but, by its authors' account, misses stable systematic bias, which is what a confidently filled region is. Trust3R [33] fits an evidential head on a frozen backbone, GeoCond [1] adds pose-level uncertainty, and Deng et al. [7] find the implied uncertainty of seven backbones at least $2.4\times$ too small. All score uncertainty against ground-truth error, which conflates confident errors on observed surfaces with confident predictions on surfaces no view saw; CUR isolates the second case. Closest to our detector, a linear probe on frozen VGGT features predicts *camera-pose* failure with AUROC 0.935 against 0.797 for native confidence [28], and a small head on the same features predicts pairwise co-visibility [34]; our support head is the dense, per-pixel analogue for unobserved *geometry*, trained either on ground-truth visibility or on the backbone's own multi-view agreement with no label at all, and the linear probe is one of our controls.

**Generative and multi-sample 3D.** Z3D [2] decodes hidden surfaces from frozen VGGT features, evidence that unobserved geometry is present in them. Feed-forward Gaussian splatting from unposed views [9] outputs pixel-aligned Gaussians with opacities, and ReconSplat [26] extrapolates beyond the observed views by design; a model built to extrapolate has all the more reason to say where it did. Real-time world models now keep a persistent 3D Gaussian memory updated from the frames they generate [4]; such a memory stores imagined and observed surfaces alike, which is exactly where a per-point support signal belongs. VolFill [20] finds sample variation $1.3\times$ larger in occluded regions; diffusion depth ensembles [11, 12] give sample-disagreement uncertainty; a generative head over quantized geometry [6, 19] whose samples disagree where the model imagines is the natural successor to our discriminative head, and future work.

**Hallucination in vision-language models.** POPE [16] labels statements unsupported by the image; ours is the geometric analogue, geometry unsupported by any view.

## 3. Defining geometric hallucination

Given input views $I_1, \ldots, I_N$, a model returns for each view $i$ a depth map $D_i$ and a confidence map $C_i$. A pixel $p$ in view $i$ is *supported* if its ground-truth surface point is visible in at least one input view, and *unsupported* if it is visible in none. A model hallucinates at $p$ when it outputs geometry at an unsupported $p$ with high confidence.

**Why confidence is the wrong tool.** Native confidence is trained as a per-pixel loss weight on observed data. It learns to be low where the regression target is noisy (edges, specular surfaces) and high where it is smooth. A region that no view sees and that the prior fills smoothly therefore receives high confidence by construction. Section 5 measures how large this effect is.

**Metrics.** All are computed per (model, scene, seed, radius, fill) and averaged with cluster-bootstrap intervals over scenes. Pixels whose confidence ties with $\tau$ count as kept, so a near-binary confidence (Pi3) is not favoured by the threshold.

- AUROC: $P(C_S > C_U)$ with ties counted half. 1.0 means every supported pixel outranks every unsupported one; 0.5 means confidence carries no information about support.
- CUR: choose $\tau$ so that 90% of supported pixels have $C \geq \tau$; CUR is the fraction of unsupported pixels with $C \geq \tau$. This is the hallucination rate you would experience after filtering by confidence.
- CUR-ERR: the fraction of unsupported pixels above $\tau$ whose AbsRel exceeds 0.1. Confident and wrong.
- $\text{AbsRel}_U$, $\text{AbsRel}_S$: depth error on unsupported and supported pixels after median scaling fitted on supported pixels only.
- Controls: the same pixels under clean input ($\text{AbsRel}_U$ ctrl, AUROC ctrl) and the ratio of median confidence inside the hidden region to its clean value.

## 4. HalluBench-3D

**Protocol.** For each scene we take 10 views. A sphere is centred on a random ground-truth surface point in the reference view, with radius 0.10 or 0.18 of the object's bounding-box diagonal, resampled until it covers 2 to 35% of the reference view. Every pixel in every view whose ground-truth point lies inside the sphere is hidden (the protocol figure in the supplement). Hidden pixels are filled with gray, a blur of the original, uniform noise, or an inpainting of the surround [27]; fill type is an ablation so that results are not an artefact of a visible patch. Labels: U is hidden in all views; S is valid and outside a 6-pixel dilation of the hidden region; the ring in between is ignored. Five mask seeds per scene; masks are deterministic in (scene, seed, radius).

**Datasets.** *DTU* [8]: the 18 scans of the MVSNet validation split, 10 views spread over the 49-camera rig, model input $420 \times 518$. *CO3Dv2* [24]: 18 scenes, the two depth-carrying sequences of each of nine categories (apple, ball, bench, hydrant, mouse, teddybear, toytrain, toytruck, vase) from the single-sequence subset, with the official COLMAP

depth as ground truth. The 10 views are spread over the full orbit, so the hidden region is seen from the back as well as the front. Frames are $640 \times 360$ phone video; models run at $518 \times 294$ (or the transpose for landscape sequences). Everything else, including every threshold, is unchanged from DTU. *ETH3D* [25]: nine high-resolution DSLR training scenes at metre to tens-of-metre scale with laser-scanner ground truth, resampled onto the undistorted pinhole frames; frames $640 \times 427$, models $518 \times 350$; the reference frame and its nine most co-visible frames. These are the building-scale scenes the first two datasets lack. *Phototourism* [10]: nine landmarks of the Image Matching Challenge 2020 set (British Museum, Florence cathedral, Lincoln Memorial, London Bridge, Milan cathedral, Mount Rushmore, Piazza San Marco, Sagrada Familia, St Paul's): Internet photographs with people and vehicles in front of the monument, unordered viewpoints and focal lengths, COLMAP multi-view-stereo depth as ground truth; per landmark the best-connected photograph and its 19 most co-visible neighbours at $640 \times 448$; all nine pass the correctness guards (supplement: ground-truth depth must reproject consistently between views and unsupported points must land inside the other views' hidden masks; agreement 0.92 to 0.999, U-transfer above 0.98). It is used only by the true-visibility protocol of Section 7, which draws its own view sets (3, 5 and 10 clean views, three rotations of the set) on every dataset. Training-data overlap: VGGT, Pi3/Pi3X and Depth Anything 3 list CO3Dv2 among their training sets [17, 29, 32]; MapAnything does not [13]; none of the four lists DTU or ETH3D. CO3Dv2 is therefore in-distribution for three of the backbones, and DTU and ETH3D are out of distribution for all of them.

**Models.** VGGT-1B [29] (depth and depth confidence), Pi3 and Pi3X [32] (local point $z$ and sigmoid confidence), MapAnything [13] (image-only metric mode), and Depth Anything 3 [17] (Large 1.1). Same views and resolution for all; fp16 on a T4. A feed-forward Gaussian-splatting model, AnySplat [9], scored on each pixel's Gaussian depth with its opacity as confidence, is in the supplement from a partial run: the representation does not change the answer.

**Statistics.** Means carry 95% cluster-bootstrap intervals over scenes (10,000 resamples; each scene contributes its mean over seeds). Pairwise comparisons use a paired sign-flip permutation test on per-scene means (20,000 permutations), Holm-corrected within each family of pairs; the smallest $p$ the test can return is $0.00005$, written $p < 0.0001$ below.

# 5. Results on HalluBench-3D

## 5.1. DTU: native confidence leaks

Table 1 gives the full DTU configuration (the per-run spread is in the supplementary material). Pooled over the two natural-statistics fills (blur and noise), CUR is 0.242 for VGGT, 0.281 for Pi3, 0.532 for Pi3X, 0.622 for MapAnything and 0.701 for Depth Anything 3, with AUROC 0.903, 0.912, 0.813, 0.743 and 0.711. Three tiers emerge. VGGT and Pi3 drop their confidence inside the hidden region (to 19 to 41% of its clean value) yet still let one in four unsupported pixels through a threshold that keeps 90% of supported ones. Pi3X leaks one in two, and under a noise fill it is as confident about geometry it cannot see as about geometry it can (ratio 1.09). MapAnything and Depth Anything 3 are close to blind to non-observation. DA3's confidence inside the hidden region stays at 80 to 86% of its clean value for every fill, including the gray patch; whatever its confidence encodes, it is not observation.

Pairwise, after Holm correction over the ten pairs: VGGT and Pi3 are indistinguishable ($p = 0.41$); both leak less than Pi3X ($p \leq 0.0015$), MapAnything and DA3 ($p = 0.0005$). Pi3X versus DA3 ($p = 0.052$), Pi3X versus MapAnything and MapAnything versus DA3 ($p = 0.19$) are not separated.

**Confident and wrong.** On DTU the imagined depth is accurate for four of the five models ($\text{ABSREL}_U$ 0.011 to 0.022, about 1.3 to 2× the clean value), so CUR-ERR stays near zero. The failure is that you cannot tell imagined from observed, not that the imagination is poor. MapAnything is the exception: 29% of its unsupported pixels are confident and wrong by more than 10% ABSREL, on top of a 10% error on supported pixels in image-only metric mode (unchanged at its native input size).

**Gate.** Before scaling the benchmark we fixed a pass condition: CUR above zero with intervals excluding zero, and AUROC clearly below one on the natural fills. It passes for all five models. The effect is not an artefact of a gray patch, and it is strongly model dependent.

## 5.2. Stress tests: the leak is not a patch artefact

A reasonable objection to Table 1 is that a filled patch is visible, and the model is reacting to the patch rather than to the absence of a surface. The stress table in the supplement answers it four ways; the numbers below are its cells.

*Fill realism.* An inpainted fill is edge-continuous and has no texture discontinuity. It raises leakage further for VGGT ($+0.088$, $p = 0.031$) and Pi3X ($+0.068$, $p = 0.005$). Table 1 is therefore a lower bound on what a natural-looking occluder produces.

*Evidence quantity.* Dropping from 10 to 3 views leaves CUR unchanged for VGGT and Pi3X ($p > 0.4$); DA3 moves

Table 1. **DTU, five models, full configuration.** 18 scans × 5 mask seeds × 2 radii × 3 fills, pooled over radii; mean [95% cluster-bootstrap CI over scans]. *conf ratio* is the median confidence inside the hidden region divided by its value under clean input. Runs P1-full and P2-full, T4, fp16.

| model | fill | AUROC | CUR | CUR-ERR | AbsRel$_U$ | AbsRel$_U$ ctrl | conf ratio |
|---|---|---|---|---|---|---|---|
| VGGT-1B | gray | 0.973 [0.955, 0.987] | 0.067 [0.031, 0.111] | 0.000 | 0.018 | 0.006 | 0.051 |
| VGGT-1B | blur | 0.886 [0.843, 0.922] | 0.263 [0.203, 0.324] | 0.000 | 0.011 | 0.006 | 0.241 |
| VGGT-1B | noise | 0.919 [0.892, 0.945] | 0.222 [0.145, 0.304] | 0.000 | 0.014 | 0.006 | 0.192 |
| Pi3 | gray | 0.981 [0.972, 0.988] | 0.053 [0.025, 0.085] | 0.001 | 0.020 | 0.014 | 0.057 |
| Pi3 | blur | 0.923 [0.895, 0.947] | 0.236 [0.156, 0.326] | 0.001 | 0.018 | 0.014 | 0.261 |
| Pi3 | noise | 0.902 [0.865, 0.934] | 0.325 [0.219, 0.438] | 0.002 | 0.020 | 0.014 | 0.411 |
| Pi3X | gray | 0.940 [0.902, 0.971] | 0.202 [0.085, 0.337] | 0.002 | 0.022 | 0.017 | 0.335 |
| Pi3X | blur | 0.826 [0.782, 0.865] | 0.512 [0.397, 0.630] | 0.002 | 0.020 | 0.017 | 0.739 |
| Pi3X | noise | 0.801 [0.734, 0.864] | 0.551 [0.390, 0.711] | 0.004 | 0.022 | 0.017 | 1.091 |
| MapAnything | gray | 0.861 [0.812, 0.905] | 0.374 [0.272, 0.480] | 0.195 [0.125, 0.269] | 0.222 | 0.111 | 0.650 |
| MapAnything | blur | 0.773 [0.720, 0.825] | 0.580 [0.482, 0.672] | 0.277 [0.197, 0.361] | 0.135 | 0.111 | 0.677 |
| MapAnything | noise | 0.712 [0.645, 0.773] | 0.664 [0.566, 0.758] | 0.295 [0.211, 0.390] | 0.129 | 0.111 | 0.709 |
| Depth Anything 3 | gray | 0.723 [0.664, 0.780] | 0.703 [0.596, 0.804] | 0.003 | 0.017 | 0.013 | 0.804 |
| Depth Anything 3 | blur | 0.682 [0.611, 0.744] | 0.711 [0.619, 0.796] | 0.003 | 0.016 | 0.013 | 0.800 |
| Depth Anything 3 | noise | 0.740 [0.686, 0.793] | 0.690 [0.578, 0.793] | 0.001 | 0.017 | 0.013 | 0.855 |

at 5 views (+0.11, $p = 0.008$) and not at 3, which we read as noise, not a trend. Having fewer views to cross-check does not make the models less sure, which says that native confidence is driven by single-view appearance rather than multi-view agreement.

*Geometry under stress.* With 3 views or low parallax, Pi3X's imagined depth degrades (AbsRel$_U$ 0.05, CUR-ERR up to 0.04) while its confidence does not move. The confident-and-wrong fraction, invisible on the easy rig, appears under harder capture.

*DA3 is different.* A distractor image raises its leakage and low-parallax input halves it, consistent with a confidence that tracks appearance consistency across views rather than observation of the surface. We flag this as an observation to explain, not as an explanation.

### 5.3. CO3Dv2: the ranking changes

Table 2 repeats the full protocol on CO3Dv2 phone captures, with every threshold frozen from DTU. Pooled over blur and noise, CUR is 0.272 for Pi3, 0.570 for Pi3X, 0.651 for VGGT, 0.797 for DA3 and 0.810 for MapAnything; AUROC is 0.891, 0.732, 0.681, 0.620 and 0.576.

The ranking changes. Pi3 is the only model whose behaviour transfers from the lab rig to phone video (CUR 0.27 on both). The leak is larger on the dataset three of the models were trained on: seeing CO3Dv2 at training time did not teach their confidence to notice a missing surface. VGGT, joint best on DTU, leaks 65% of unsupported pixels on CO3Dv2, 2.7× its DTU value, and under a noise fill its confidence is almost blind to non-observation (AUROC 0.59). MapAnything and DA3 let about four in five unsupported pixels through. After Holm correction, Pi3 leaks less than every other model ($p = 0.0005$ each); Pi3X leaks less than MapAnything and DA3 ($p = 0.0005$); VGGT leaks less than MapAnything ($p = 0.010$) and DA3 ($p = 0.0005$); VGGT versus Pi3X ($p = 0.23$) and MapAnything versus DA3 ($p = 0.73$) are not separated. Across datasets, the intervals of VGGT, MapAnything and DA3 do not overlap; those of Pi3 and Pi3X do.

**Imagination is accurate here too.** AbsRel$_U$ is within 0.005 of AbsRel$_S$ for every model, so CUR-ERR stays below 0.04, and MapAnything's confident-and-wrong behaviour from DTU does not appear (AbsRel$_S$ 0.041, CUR-ERR 0.007). These are everyday objects with strong shape priors; the models complete them well and report the completion with the confidence of an observation.

**Fill effects differ from DTU.** On DTU every fill raised CUR over gray for VGGT, Pi3 and Pi3X. On CO3Dv2 noise raises VGGT's CUR by 0.20 ($p < 0.0001$) but blur does not ($p = 0.30$); Pi3X reacts to both (+0.26, +0.23); Pi3 weakly (+0.06, +0.10); MapAnything and DA3 not at all ($p$ from 0.06 to 0.91). A confidence that does not change when the patch changes is not reading the patch.

**Per-scene view.** Figure S3 in the supplementary shows that the leak is not driven by a few scenes. Removing the one scene that missed the mask-transfer guard moves every pooled number by at most 0.013.

### 5.4. Cheap detectors do not help

Two obvious alternatives, scored on the Table 1 rows with blur, radius 0.18 and seed 0 (table in the supplement). Variance over $K = 4$ photometrically perturbed runs is much worse than native confidence on both backbones (AUROC 0.67 and 0.55 against 0.92 and 0.83, paired $p \leq 0.0007$) while costing four forwards: the imagined geometry is stable under input perturbation, the systematic bias ReVar3R's authors describe. Reprojection consistency with the model's own cameras is no better than chance (AUROC 0.49, 0.48):

Table 2. **CO3Dv2, five models, full configuration.** 18 scenes × 5 seeds × 2 radii × 3 fills, pooled over radii; mean [95% CI over scenes]. Runs P2-co3d and P2-co3d-b.

| model | fill | AUROC | CUR | CUR-ERR | AbsRel$_S$ | AbsRel$_U$ | AbsRel$_U$ ctrl | conf ratio |
|---|---|---|---|---|---|---|---|---|
| VGGT-1B | gray | 0.747 [0.692, 0.802] | 0.565 [0.453, 0.677] | 0.029 | 0.029 | 0.031 | 0.027 | 0.634 |
| VGGT-1B | blur | 0.772 [0.717, 0.825] | 0.534 [0.416, 0.652] | 0.034 | 0.029 | 0.031 | 0.027 | 0.603 |
| VGGT-1B | noise | 0.590 [0.517, 0.656] | 0.767 [0.693, 0.839] | 0.029 | 0.029 | 0.029 | 0.027 | 0.805 |
| Pi3 | gray | 0.930 [0.894, 0.958] | 0.196 [0.118, 0.283] | 0.002 | 0.015 | 0.016 | 0.013 | 0.351 |
| Pi3 | blur | 0.903 [0.860, 0.939] | 0.251 [0.170, 0.334] | 0.002 | 0.015 | 0.016 | 0.013 | 0.389 |
| Pi3 | noise | 0.880 [0.813, 0.929] | 0.293 [0.193, 0.405] | 0.002 | 0.015 | 0.015 | 0.013 | 0.472 |
| Pi3X | gray | 0.863 [0.825, 0.899] | 0.329 [0.248, 0.408] | 0.001 | 0.017 | 0.022 | 0.014 | 0.749 |
| Pi3X | blur | 0.720 [0.661, 0.777] | 0.584 [0.500, 0.668] | 0.002 | 0.017 | 0.018 | 0.014 | 0.877 |
| Pi3X | noise | 0.743 [0.680, 0.798] | 0.556 [0.462, 0.650] | 0.001 | 0.017 | 0.019 | 0.014 | 0.867 |
| MapAnything | gray | 0.609 [0.554, 0.665] | 0.786 [0.721, 0.849] | 0.004 | 0.041 | 0.029 | 0.034 | 0.632 |
| MapAnything | blur | 0.575 [0.514, 0.637] | 0.801 [0.735, 0.863] | 0.008 | 0.042 | 0.028 | 0.034 | 0.674 |
| MapAnything | noise | 0.578 [0.524, 0.631] | 0.819 [0.765, 0.870] | 0.007 | 0.041 | 0.033 | 0.034 | 0.690 |
| Depth Anything 3 | gray | 0.619 [0.560, 0.674] | 0.816 [0.751, 0.877] | 0.001 | 0.019 | 0.015 | 0.014 | 0.874 |
| Depth Anything 3 | blur | 0.622 [0.574, 0.668] | 0.813 [0.748, 0.872] | 0.001 | 0.018 | 0.015 | 0.014 | 0.853 |
| Depth Anything 3 | noise | 0.618 [0.553, 0.679] | 0.780 [0.702, 0.854] | 0.000 | 0.019 | 0.016 | 0.014 | 0.901 |

the imagined surface is cross-view coherent. MapAnything's multi-view consistency confidence leaks more than its learned confidence (AUROC 0.67, CUR 0.78 against 0.74 and 0.62 on the same rows, run P6-mapanything-mv).

## 6. Method: observe, imagine, abstain

The benchmark says that native confidence does not know what the model has seen. Rather than suppress the prior, which is what makes these models useful, we want the model to say which of its outputs came from the prior.

**Architecture.** A frozen backbone (VGGT, Pi3X, Pi3 or Depth Anything 3) provides geometry, native confidence, and the multi-view features that produced them: for VGGT and DA3 the tokens of the four cross-view layers their depth heads read (2048-d at $14 \times 14$ patches), for Pi3X and Pi3 the 2048-d decoder tokens feeding their confidence heads. The *support head* (LayerNorm and $1\times1$ projection per layer to 256 channels, sum, two $3\times3$ convolutions at patch resolution, one logit per patch upsampled bilinearly) outputs $P(\text{imagined})$; the benchmark scores $1 - P(\text{imagined})$. 3.3 M parameters on VGGT and DA3, 1.7 M on Pi3X and Pi3; the backbone is neither modified nor re-run.

**Training labels.** Three label sources, all on the 79 scans of the DTU training split, disjoint from the 18 benchmark scans and the 22 extension scans of Section 7; the views are clean, 3, 5 or 10 at random per step. *Self-support labels* (the main head, label-free): the backbone is run once; each pixel's *predicted* point is projected with the *predicted* depth and cameras into the other input views and counted as supported where it lands on their predicted surface (3% depth tolerance, the 20% least confident pixels dropped). A pixel no other view supports is the positive class, a pixel supported by two or more views outside a 3-pixel margin the negative class. This is the reprojection test of Section 5.4 used as a teacher, so the head has to generalise beyond it to be worth having. *Visibility labels* (the upper bound): the same count from ground-truth depth and cameras. *Fill labels* (the cautionary variant): a sphere placed on the predicted surface is hidden in every view (gray, blur or noise; inpaint held out) and the head learns to find the hidden pixels, with 15% clean samples. All three: class-balanced binary cross-entropy, 1500 AdamW steps (learning rate $10^{-3}$, cosine, warm-up 100), one scene of random views per step, 10 minutes per backbone on one H200 (1.2 hours on a T4); hyperparameters fixed before any benchmark scan was evaluated.

**Inference and compute.** One backbone forward, one head forward; the head adds under 1% of the backbone's FLOPs. Three colours for the user: *observed* above $1 - P(\text{imagined}) = 0.7$, *imagined* below 0.3, *abstain* between, fixed in advance; the metrics use the continuous score.

**Controls and ablations.** A head could learn an image cue (the patch; image borders and depth edges, where single-view pixels concentrate) rather than the missing support. The decisive control is the *per-image feature head*: the same head on per-image encoder tokens before any cross-view attention, trained on the same labels. Two hand-crafted scores test the same objection without training: the distance of a pixel to the image border and minus the gradient of the predicted depth. Further controls: a *linear probe* on the same features [28] and on each cross-view layer alone, two extra initialisation seeds, the held-out inpainted fill and CO3Dv2 for the fill-trained head; heads sharing a label source train jointly on the same backbone forwards.

## 7. Experiments on the method

**Fill-trained head: what it learns.** Its pre-registered test asked it to beat native confidence ($p < 0.05$) and to beat its per-image control on the held-out inpainted fill and on CO3Dv2. It scores AUROC 0.990 to 1.000 on every cell of the full configuration and the stress tests, and so does the control: 1.000 against 0.999 (VGGT) and 0.994 against 0.994 (Pi3X) on the inpainted fill, 0.992 against 0.989 and 0.969 against 0.988 on CO3Dv2. Its formal clause passed on a 0.003 difference, a flaw of the criterion; put to real single-view geometry (supplement table), the same head scores *below* native confidence on every dataset. It learned the fill and nothing else; the per-image control is mandatory for anyone training on masking labels.

**What the tokens know.** The per-image control falls below chance on CO3Dv2 and ETH3D, and so does the distance to the image border (indistinguishable from the control there, $p > 0.3$): a head that cannot see across views learns the frame edge, which marks single-view pixels on the rig and not elsewhere. A linear probe on the same cross-view features scores within 0.015 of the convolutional head of the next paragraphs on every dataset, so the information is in the features and any head is a reader. A probe on one VGGT layer at a time says where it appears (DTU / CO3Dv2): layer 4 of 24 scores 0.87 / 0.48, like the per-image control; layer 11 0.95 / 0.70; layer 17 0.99 / 0.92; layer 23 0.99 / 0.88. Support becomes linearly decodable in the second half of the aggregator; the early layers carry only the per-image cue.

**Visibility head: the pre-registered test.** Gate 3 was written before any visibility head was trained, with a margin this time: beat native confidence on AUROC and CUR ($p < 0.05$) and the per-image control on AUROC by at least 0.02 ($p < 0.05$), on DTU and on one unseen dataset. Table 3: the ground-truth-trained head reaches AUROC 0.990 to 0.993 on DTU against 0.77 to 0.84 for native confidence and 0.86 to 0.88 for the control, $+0.11$ to $+0.13$ at every view count; off the rig it keeps 0.93 to 0.98 while the control falls *below chance* on CO3Dv2. Every paired difference is at the permutation floor ($p < 0.0001$ on 18 scenes, $0.0043$ on nine). Gate 3 passes on three datasets and three backbones.

**Label-free head: the method.** Gate 4, written before any label-free head was trained, asks the same of the head trained on the backbone's own support count, and adds: within 0.03 AUROC of the ground-truth-trained head on DTU, and ahead of reprojection consistency, its own teacher, on CO3Dv2 at $p < 0.05$. It passes on all three backbones of Table 3, and on Depth Anything 3, the worst leaker of Table 1, where the label-free head scores 0.98 / 0.88 / 0.97 / 0.93 on DTU / CO3Dv2 / ETH3D / landmarks against 0.84 / 0.68 / 0.65 / 0.78 for native confidence and 0.76 / 0.44 / 0.75 / 0.73 for its per-image control (supplement). The label-free head is within 0.02 of the ground-truth-trained head on every dataset, beats its own per-image control by 0.10 to 0.12 on DTU and by 0.35 to 0.50 elsewhere, and beats reprojection consistency by 0.04 to 0.07 on DTU and by 0.14, 0.05 and 0.04 on CO3Dv2 ($p < 0.0001$, $0.006$, $0.008$). On ETH3D the two tie ($\pm 0.02$, $p > 0.15$): with adjacent DSLR views of a building the reprojection test is as good as any head, and the head's advantage there is cost alone.

**Tourist photographs, adaptation, more views.** The landmark photographs are the capture the detector is meant for and the hardest for the alternatives: people stand between camera and monument and focal lengths differ from photo to photo, so the model's own cameras are least reliable and reprojection consistency scores only 0.81 to 0.82. The label-free head scores 0.93 to 0.94 on all three backbones against 0.80 to 0.85 for native confidence (paired $p \leq 0.016$) and 0.72 to 0.73 for the per-image control; it beats reprojection consistency by 0.11 to 0.13 ($p = 0.0043$, the nine-scene floor). Trained on the target capture itself, still without labels (one half of a dataset, scored on the other), the head gains where the backbone's own geometry on that capture is reliable, Pi3X on CO3Dv2 ($+0.025$, $p = 0.007$) and on ETH3D ($+0.027$, $p = 0.008$), changes nothing on the landmarks ($\pm 0.01$, $p > 0.4$), and hurts VGGT on CO3Dv2 ($-0.09$, $p = 0.0004$), whose reprojection test, the teacher, scores 0.79 there against 0.89 for Pi3X. The DTU-trained head is the safe default and never touches the target capture.

**What abstaining buys.** CUR sets its threshold on the test scene itself, which a deployed filter cannot do. From the per-pixel scores (run P6-dump; table in the supplement): the area under the single-view rate of the kept pixels against coverage is within 0.002 to 0.02 of the oracle for the label-free head on every dataset and two to four times larger for native confidence; keeping the half the head trusts most lowers the depth error of what remains by 15 to 44% (native confidence lowers it as much: it tracks error, the head tracks support). With the bands 0.3 and 0.7 fixed before any evaluation, the head marks 57 to 87% of the labelled pixels *observed*, of which 0.5 to 5% are single-view (per-image control: 50 to 94% with 4 to 20%). Downstream, fusing the views with the backbone's own cameras and keeping the 80% of points the head trusts most raises the accuracy of the fused cloud on DTU from 0.65 to 0.77 (native confidence 0.78, reprojection 0.76, per-image control 0.70, random 0.65), while the oracle that drops single-view pixels first reaches 0.66: on DTU single-view geometry is nearly as accurate as supported geometry, so abstaining buys corroboration, not accuracy, and native confidence, which tracks error, filters accuracy as well (ETH3D is the same picture; supplement). A threshold

Table 3. **True-visibility protocol: detecting single-view geometry from clean views.** Pooled AUROC and CUR over scenes and view counts (3, 5 and 10 views; 3 and 5 on CO3Dv2, where ten views around an object leave no single-view pixel). No head saw any of these scenes; the last row per backbone is the label-free head trained on the other half of the target dataset instead of DTU. Runs P4-vis*, P7-vis-imc, P6-adapt-*, P7-adapt-imc. DA3 and MapAnything native confidence on the same rows: AUROC 0.836 / 0.675 / 0.645 / 0.782 and 0.848 / 0.719 / 0.789 / 0.893. Intervals, per-view-count tables and paired tests: supplement.

| detector | DTU | | CO3Dv2 | | ETH3D | | Photot. | |
|---|---|---|---|---|---|---|---|---|
| | AUROC | CUR | AUROC | CUR | AUROC | CUR | AUROC | CUR |
| *VGGT-1B* | | | | | | | | |
| native confidence | 0.772 | 0.530 | 0.754 | 0.723 | 0.715 | 0.566 | 0.802 | 0.451 |
| reprojection consistency | 0.954 | 0.082 | 0.793 | 0.693 | 0.975 | 0.069 | 0.824 | 0.340 |
| distance to image border (any backbone) | 0.701 | 0.727 | 0.454 | 0.897 | 0.725 | 0.632 | 0.730 | 0.649 |
| depth edges (any backbone) | 0.549 | 0.831 | 0.422 | 0.943 | 0.598 | 0.762 | 0.674 | 0.619 |
| per-image control (GT labels) | 0.878 | 0.340 | 0.462 | 0.878 | 0.712 | 0.666 | 0.787 | 0.530 |
| support head, GT visibility labels | 0.993 | 0.015 | 0.927 | 0.211 | 0.980 | 0.055 | 0.908 | 0.258 |
| per-image control (self labels) | 0.885 | 0.325 | 0.460 | 0.892 | 0.626 | 0.750 | 0.729 | 0.681 |
| **support head, label-free** | 0.992 | 0.017 | 0.931 | 0.199 | 0.980 | 0.051 | 0.930 | 0.191 |
| label-free head, adapted on the target domain | – | – | 0.841 | 0.474 | – | – | 0.926 | 0.203 |
| *Pi3X* | | | | | | | | |
| native confidence | 0.820 | 0.493 | 0.697 | 0.846 | 0.887 | 0.304 | 0.839 | 0.450 |
| reprojection consistency | 0.914 | 0.164 | 0.885 | 0.224 | 0.977 | 0.054 | 0.814 | 0.361 |
| per-image control (GT labels) | 0.863 | 0.375 | 0.428 | 0.889 | 0.694 | 0.685 | 0.775 | 0.577 |
| support head, GT visibility labels | 0.990 | 0.021 | 0.930 | 0.221 | 0.975 | 0.072 | 0.938 | 0.176 |
| per-image control (self labels) | 0.871 | 0.363 | 0.469 | 0.877 | 0.610 | 0.776 | 0.719 | 0.697 |
| **support head, label-free** | 0.988 | 0.025 | 0.939 | 0.190 | 0.963 | 0.103 | 0.940 | 0.166 |
| label-free head, adapted on the target domain | – | – | 0.964 | 0.094 | – | – | 0.929 | 0.198 |
| *Pi3* | | | | | | | | |
| native confidence | 0.841 | 0.489 | 0.551 | 0.882 | 0.814 | 0.498 | 0.847 | 0.447 |
| reprojection consistency | 0.952 | 0.089 | 0.894 | 0.205 | 0.976 | 0.039 | 0.817 | 0.345 |
| per-image control (GT labels) | 0.878 | 0.345 | 0.464 | 0.880 | 0.717 | 0.661 | 0.796 | 0.510 |
| support head, GT visibility labels | 0.992 | 0.016 | 0.934 | 0.203 | 0.953 | 0.135 | 0.932 | 0.201 |
| per-image control (self labels) | 0.888 | 0.320 | 0.471 | 0.895 | 0.654 | 0.750 | 0.724 | 0.683 |
| **support head, label-free** | 0.989 | 0.021 | 0.930 | 0.225 | 0.951 | 0.139 | 0.931 | 0.181 |

set to keep 90% of the supported pixels of the *other* scenes of a dataset lets 1 to 2% of single-view pixels through on DTU, 7 to 16% on CO3Dv2 and ETH3D and 17 to 20% on the landmarks, against 57 to 83% for native confidence; a DTU threshold applied unchanged elsewhere keeps 57 to 92% of the supported pixels at single-view rates of 0.6 to 19%, whereas VGGT's native confidence has no transferable scale.

**The benchmark's own question.** The benchmark defines unsupported geometry as a surface no input view shows; the vis protocol's positive class is a surface only one view shows. Put back to the benchmark's filled patches (run P6-vis-on-fill), the visibility heads score AUROC 0.70 to 0.75 on DTU and 0.62 to 0.67 on CO3Dv2, the controls 0.60 to 0.65, border and depth edges 0.43 to 0.51, all below native confidence on the same rows (run P6-vis-on-fill). A head trained to find single-view geometry does not fire on a patch every view shows, for the reason the reprojection and perturbation baselines do not: the invented surface is coherent across views and across the features. It is coherent across models too. Disagreement between two backbones on the same views (depths aligned by a label-free median ratio, six pairs over VGGT, Pi3X, Pi3, DA3 and MapAnything; run P8-xmodel) scores AUROC 0.52 to 0.61 on the DTU patches and 0.49 to 0.63 on CO3Dv2 (supplement): five models of three families fill the hole with the same surface. The head certifies agreement between views, and a consistent invention has it. The detector answers "which pixels lack corroboration", not "which surfaces were invented".

## 8. Limitations and conclusion

The head certifies multi-view agreement, not truth: a surface every view agrees on passes, whether a camera saw it or the prior drew it, and the benchmark's filled patch is that case. The benchmark labels are a proxy for real occlusion, and the stress tests bound but do not remove that gap. CO3Dv2 ground truth is COLMAP depth with 1 to 3% noise, and three backbones saw CO3Dv2 in training. Ground truth on the landmarks is multi-view stereo with holes, so single-view labels there are a lower bound. MapAnything has only its native confidence on the vis protocol. Static scenes only. All findings depend on the released checkpoints.

When a feed-forward 3D model has not seen a surface, it mostly does not say so: a quarter to four fifths of the geometry it invents passes a confidence filter tuned to keep 90% of what it observed. The answer is not to stop the models from imagining but to make them say when they do. The models already know: a linear probe on their cross-view features separates single-view from supported geometry at AUROC 0.99 where native confidence scores 0.8, and a small head trained on the backbone's own multi-view agreement, with no ground truth, reads the same signal on four backbones and four datasets it never saw. What it cannot do sets the next problem: a surface the model invents consistently in every view passes every test we have, and five different models invent the same one.

## LLM Usage Statement

A general-purpose LLM assistant was used, under the authors' direction, for parts of the code (the benchmark runner, the model wrappers, the run scripts and figure scripts), for literature lookup, and for editing the text of this paper, including grammar. The study design, the definitions and metrics, the choice and freezing of every hyperparameter, the execution and checking of every experiment, the verification of every reported number against its logged run, and the conclusions are the authors' own, and the authors are responsible for all content. The assistant was not used to generate results, figures from unlogged data, or references; every citation was checked against the published record. The tool is named in the camera-ready version.

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